$A+(-A) = [a_1\ a_2 \ldots\ldots a_n] + [-a_1\ -a_2 \ldots\ldots -a_n]$

$= [a_1+(-a_1)\ \ a_2+(-a_2) \ldots\ldots a_n+(-a_n)]$ (Vector-matrix addition)

$= [0\ 0 \ldots\ldots 0]$ (Algebraic property (iv))

$= 0$

$C^{-1}((A+(-A))+CB)B^{-1} = C^{-1}(0+CB)B^{-1} = C^{-1}(CB+0)B^{-1}$ (Theorem 1(a))

$= C^{-1}(CB)B^{-1}$ (Theorem 1(c))

$= ((C^{-1}C)B)B^{-1}$ (Theorem 2(a)) $= (IB)B^{-1}$ (By the defn. of inverse)

$= BB^{-1}$ (Theorem 2(e)) $= I$ (By the defn. of inverse) $\square$